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Skills Should Not Depend on Who Can Pay

A conditions-of-access framework for digital learning in Kenya and Sub-Saharan Africa: connectivity, affordability, inclusion and instructional quality

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HIGHLIGHTS

- Access to digital learning is modelled as four conditions that can fail independently: connect, afford, use and learn.
- In Sub-Saharan Africa about 60% of people live under mobile broadband coverage but do not use it, against a 13% coverage gap.
- Mobile data costs about 5.8% of monthly income in Africa against a 2% affordability target, and women are 32% less likely than men to use mobile internet.
- Experimental evidence shows that devices without instruction yield little learning, whereas targeted instruction and low-technology designs can.
- Seven design requirements and six testable propositions are derived for payer-independent access.

ABSTRACT

Purpose: Digital learning is promoted as a way around shortages of teachers, classrooms and money, yet the conditions under which it widens rather than narrows inequality are rarely stated. This paper develops and tests, against the published evidence, a conditions-of-access framework for Kenya and Sub-Saharan Africa.

Design: A theory-building narrative review draws on international agency statistics, systematic reviews and randomised or quasi-experimental studies, interpreted through the capability approach and the digital-divide literature. Findings: Access fails at four sequential and cumulative gates: connection, affordability, use (language, skills, gender, disability) and learning. The binding constraint is increasingly use and cost, not coverage: in 2023, 27% of the Sub-Saharan population used mobile internet, 60% lived under coverage without using it, and 13% lived outside coverage. Data cost 5.8% of monthly income against a 2% target, women were 32% less likely than men to use mobile internet, and children with disabilities were 2.5 times more likely never to attend school. Technology improves learning when it delivers instruction matched to the learner's level and fails when devices are distributed alone. Implications: Seven design requirements and six propositions are derived for providers who wish access to be independent of the learner's ability to pay, together with a measurement agenda centred on the learner who is hardest to reach. Originality: The paper integrates the capability approach, the multi-level digital divide and experimental learning evidence into one operational framework, and applies it reflexively to a Kenyan provider.

Keywords: digital learning; access to education; digital divide; capability approach; affordability; inclusive design; Kenya; Sub-Saharan Africa

1. Introduction

1.1 The promise and the problem

The promise of digital learning in low- and middle-income countries rests on a simple economic idea. If content can be copied at near-zero marginal cost, then skills need not be rationed by the number of teachers, the size of classrooms or the size of a family's budget. The idea attracts governments, funders and providers because it appears to break the link between what people can pay and what they can learn. It is also incomplete. A course that can be copied freely must still reach a person, in a form that person can use, at a price that person can bear, and it must teach something. Each of these requirements can fail, and the failures are not randomly distributed. They fall most heavily on the learners the technology is meant to serve: the poor, women, people with disabilities, and people in rural areas or outside the dominant language community.

The scale of the underlying need is large. The 2022 update of the State of Global Learning Poverty, produced by the World Bank, UNESCO, UNICEF, the United Kingdom's Foreign, Commonwealth and Development Office, USAID and the Bill & Melinda Gates Foundation, estimated that about 70% of ten-year-olds in low- and middle-income countries could not read and understand a simple text, up from 57% before the pandemic, and projected the figure for Sub-Saharan Africa at 89% (World Bank et al., 2022). The labour-market consequences are

visible in Kenya's youth cohort. Figures from the Kenya National Bureau of Statistics, reported by the Business Daily, show about 3.5 million people aged 15 to 34, roughly 19% of that age group, neither employed nor in education or training (Business Daily, 2023). A skills market that reaches only those who can already pay does not touch this population, and may widen the gap between them and their better-resourced peers.

1.2 The gap in the literature

Three literatures bear on the problem, and they are seldom brought together. The first, rooted in development economics and philosophy, treats access as a question of capability: what people are actually able to do and be, given the resources and conditions they face (Sen, 1999; Robeyns, 2005). The second, from communication and media studies, shows that the digital divide is not one gap but several, from motivational and material access through skills to the tangible outcomes of use (van Dijk, 2005; van Deursen & van Dijk, 2019). The third, from education economics, uses randomised and quasi-experimental designs to ask what raises learning, and finds that technology helps when it carries good instruction and does not when it replaces it (Escueta et al., 2020; Muralidharan et al., 2019; UNESCO, 2023). Each addresses part of the problem. The capability literature is rich in principle but rarely operational. The digital-divide literature rarely measures learning. The experimental literature measures learning but gives little attention to who is excluded before the experiment begins. A provider who must decide what to build, what to charge and whom to reach needs the three combined.

1.3 Contribution and structure

This paper makes three contributions. First, it proposes a conditions-of-access framework with four gates, connect, afford, use and learn, that can be applied to any digital learning programme and that locates where a particular learner is lost. Second, it reviews the evidence for Kenya and Sub-Saharan Africa gate by gate and derives seven design requirements and six testable propositions. Third, it applies the framework reflexively to a Kenyan provider, stating honestly where that provider meets the requirements and where it does not.

The paper proceeds as follows. Section 2 describes the review method and its limits. Section 3 develops the theoretical foundations and the framework. Sections 4 to 7 review the evidence on connection and affordability, on use, on learning, and on the funding of access. Section 8 states the propositions. Section 9 derives design requirements. Section 10 applies the framework to Instracta. Section 11 discusses implications for policy, practice, funders and research. Section 12 sets out limitations, and Section 13 concludes. The first author is the founder of the provider discussed in Section 10; this interest is declared in the closing statements and the reader should weigh the argument accordingly.

2. Method

2.1 Type of review

This is a theory-building narrative review. It is not a systematic review conducted to a PRISMA protocol, and it reports no screening counts. The purpose is to assemble the strongest available evidence on each condition of access, to integrate it into a framework, and to derive propositions that can be tested. Narrative and conceptual reviews are an accepted form in education and development journals when the aim is synthesis and theory development rather than the estimation of a pooled effect. Their principal weakness is selection: without a protocol, a reviewer can choose evidence that supports a preferred conclusion. The steps below are intended to limit that risk and to allow a reader to reproduce the search.

2.2 Search strategy and sources

Searches were conducted in October 2026 using a general web search engine and repository search across publisher and open-access indexes (including the records of the American Economic Association, the National Bureau of Economic Research, the World Bank Open Knowledge Repository, EdTech Hub and Open Development and Education, and IDEAS/RePEc). Search strings combined terms for the four conditions with regional and design terms, for example mobile internet usage gap Sub-Saharan Africa, 1 GB affordability target, children with disabilities out of school, learning poverty, technology-aided instruction randomised, and low-tech learning school closures. Appendix A lists the principal strings. International agency reports were retrieved from the issuing bodies (GSMA, ITU, A4AI, UNESCO, WHO, UNICEF and the World Bank) or from reputable secondary reports of them, which are marked in the text.

2.3 Inclusion, appraisal and synthesis

Sources were included if they met at least one of three criteria: they reported population-level statistics from an agency with a mandate and method for collecting them; they reported experimental or quasi-experimental estimates of learning effects in low- or middle-income countries; or they made a substantive theoretical contribution to the study of access, capability or the digital divide. Vendor reports, blogs and promotional material were not used as evidence, although news reports were used to locate primary statistics, and this is flagged. Appraisal used a simple hierarchy: meta-analyses and systematic reviews, then randomised and quasi-experimental studies, then descriptive statistics from agencies, then theory. Effect sizes are reported as the authors state them, in standard deviations, and are not pooled, because outcomes, populations and time horizons differ.

Synthesis proceeded by assigning each finding to one of the four gates, comparing convergent and divergent results within each gate, and then asking what each gate implies for design. Where the Kenyan evidence is thin, which is often, the paper says so and relies on regional or

international evidence with an explicit caveat. Several sources could be accessed only through abstracts or catalogue records, and statements about them are confined to what those records report.

2.4 Positionality

The first author is a practitioner: an instructional designer, learning management system lead and founder of a Kenyan learning provider. Practitioner knowledge informed the choice of questions and the design requirements. It also creates an interest in the conclusions. The paper therefore states its claims in a form that can be checked against sources and presents counter-evidence where it exists.

3. Theoretical foundations and the conditions-of-access framework

3.1 Access as capability

The capability approach asks what people are able to do and to be. Sen (1999) argued that the proper measure of development is the expansion of the real freedoms that people enjoy, and that resources matter only through what people can make of them. The same resource can yield very different capabilities for different people, because of personal characteristics, social conditions and environment, which Sen and later writers call conversion factors (Robeyns, 2005). Applied to learning, a platform is a resource. A learner who completes a course and gains a skill she can use has realised a capability. The distance between the two is the substance of the access problem, and it is a distance that conversion factors such as language, disability, gender norms, electricity, time and money determine.

The approach has been applied to information and communication technologies for development. Kleine (2013) used it to argue that evaluations of ICT projects should ask whether technology expands people's real choices, not only whether it is adopted. Its practical value for this paper is to shift attention from the average user to the person with the fewest conversion factors in her favour. It does not tell a provider what to build. The next two literatures help to do so.

3.2 The multi-level digital divide

Early discussions of the digital divide treated it as a binary between those with and without a connection. Subsequent work has distinguished levels. The first level concerns physical and material access: devices, connections and the ongoing costs of hardware, software and subscriptions. The second concerns skills and uses. The third concerns the tangible outcomes that different groups obtain from use (van Dijk, 2005). The argument that physical access is

giving way to material access is made by van Deursen and van Dijk (2019), who found in a representative Dutch survey that the range of devices people own, their device-related opportunities and the continuing costs of use shape inequalities in skills, uses and outcomes. The finding comes from a high-income country but its logic travels: owning a connection is different from being able to afford to use it for learning.

A corollary is important for African contexts. If the first-level divide is already a matter of material access in the Netherlands, it is more so where handset costs and data prices are high relative to income. The evidence in Section 4 shows that this is so.

3.3 Why the experimental evidence on learning belongs in a theory of access

Access theories usually stop at use. The argument of this paper is that they should continue to learning, for two reasons. The first is normative: the purpose of access to education is learning, and an access that delivers none is not access to education in a meaningful sense. The second is empirical: a large body of evidence shows that the same technology yields very different learning depending on how it is used (Escueta et al., 2020; UNESCO, 2023). A theory of access that is silent on instructional quality will therefore overstate what connection and devices achieve. The Global Education Monitoring Report of 2023 states the point in policy terms: the focus should be on learning outcomes, not digital inputs (UNESCO, 2023).

3.4 The framework: four gates

We combine the three literatures in a framework with four conditions, or gates, that a learner must pass to realise a capability through digital learning. Figure 1 depicts them.

1. Connect. The learner lives within reach of the infrastructure the service needs, such as a mobile network, electricity, or a physical learning point.
2. Afford. The learner, or someone acting for the learner, can pay the full cost of using the service, including data, devices and time, and not only the course fee.
3. Use. The learner can operate the service given language, literacy, digital skills, gender norms and any disability. This is the gate addressed by inclusive and universal design.
4. Learn. The experience leads to learning gains that are measurable and relevant. Reaching a learner with poor instruction is not access to education.

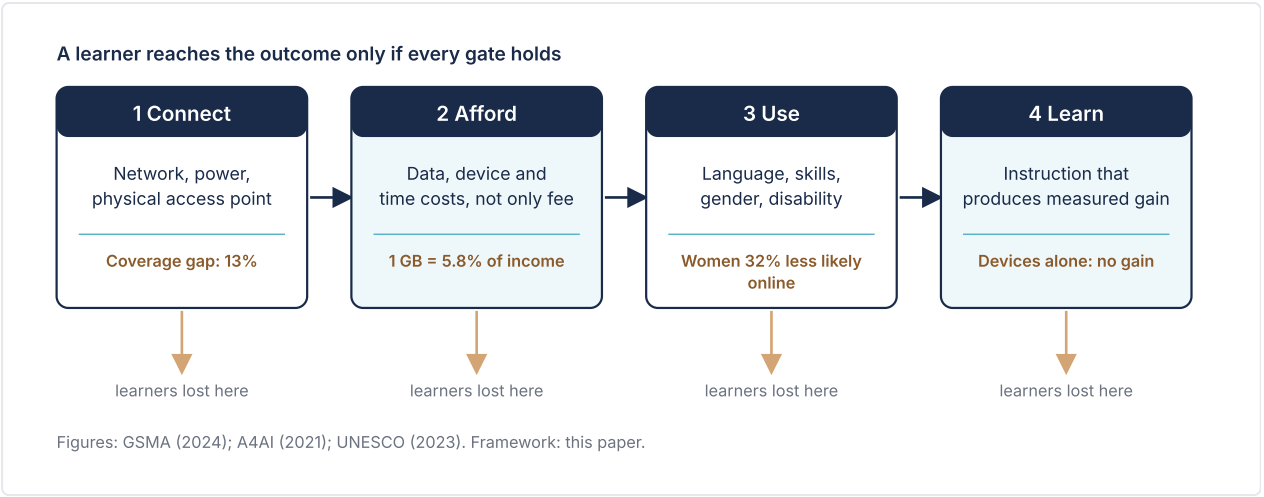


Figure 1. Conditions-of-access framework. Four gates (connect, afford, use, learn) determine whether a learner realises the capability that a digital learning programme is meant to provide. Examples beneath each gate are indicative evidence reviewed in this paper.

The gates are sequential in one sense and cumulative in another. A learner who cannot connect never reaches the later gates. A provider who solves three and neglects one still excludes. The framework is deliberately modest. Its purpose is to let a funder, a regulator or a provider ask, of any programme, at which gate a given learner is lost, and what it would cost to open it. It is also a diagnostic of where the evidence is thin: the gates differ in how well they are measured, and the least measured, use and learning among the excluded, are those that matter most.

3.5 Relation to the companion papers

This paper is the first of three working papers, each addressing one of the principles on which Instracta's work is organised: access, evidence and local provision. The second paper develops a proportionate evidence standard for judging whether a programme works, which is the standard to which the learn gate is held. The third paper addresses language, pedagogy and institutions, which shape the use gate and the sustainability of any solution. The three are designed to be read together. A programme that is accessible but unevaluated cannot say whether it teaches, and one that is evaluated but built in a language or pedagogy that does not fit will not scale.

4. Gates 1 and 2: connection and affordability

4.1 Coverage is no longer the main gap

The most important pattern in the connectivity data is that coverage is no longer the dominant constraint. The GSMA's connectivity series distinguishes a coverage gap, people who live

outside mobile broadband coverage, from a usage gap, people who live within coverage but do not use mobile internet. In the 2024 edition, reported at the end of 2023, 27% of the Sub-Saharan African population used mobile internet, 60% lived within coverage but did not use it, and 13% lived outside coverage (GSMA, 2024, as reported by Connecting Africa, 2024). For every person outside coverage, there are more than four people under coverage who are not online. The previous edition had reported that a quarter of the regional population used mobile internet at the end of 2022 (GSMA, 2023). The International Telecommunication Union estimated that about 2.6 billion people, 33% of the world's population, were offline in 2023, and that 37% of people in Africa and 27% in low-income countries used the internet (ITU, 2023).

The implication is that building towers is not the primary task. Making use possible and worthwhile is. A provider that assumes that learners are 'unconnected' and waits for infrastructure misreads the situation, as does a provider that assumes that coverage implies use. Figure 2 summarises the connectivity, cost and gender indicators.

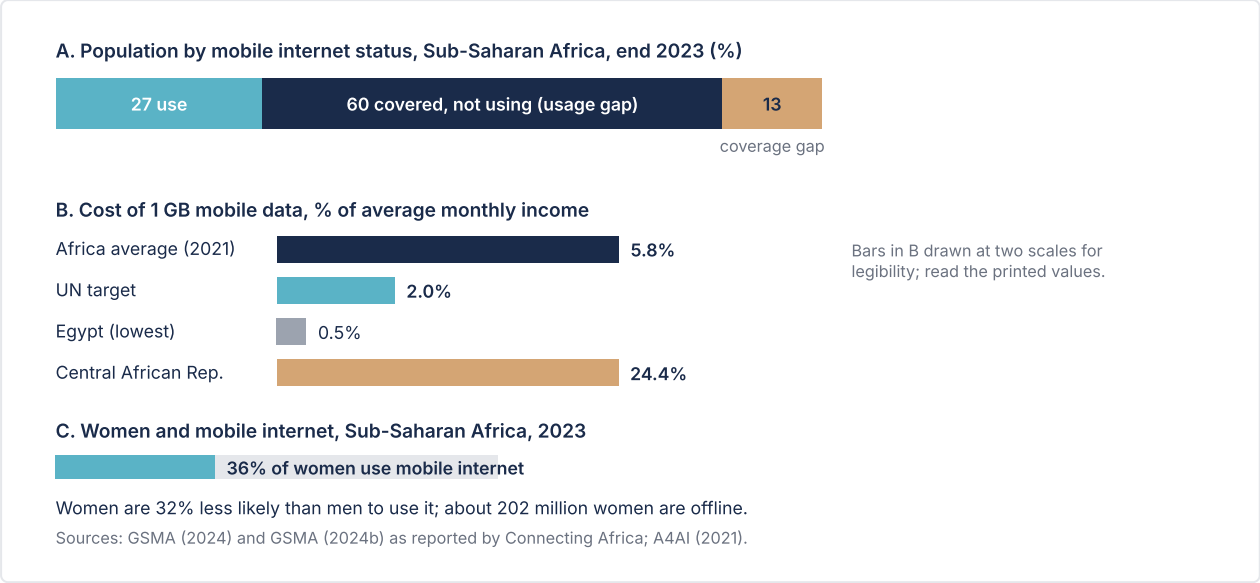


Figure 2. Connectivity, cost and gender indicators for Sub-Saharan Africa. Panel A shows the share of the population using mobile internet, covered but not using it, and outside coverage (end 2023). Panel B shows the cost of 1 GB of mobile data as a share of average monthly income against the UN target. Panel C shows women's use of mobile internet in 2023.

4.2 Cost is a principal reason for non-use

The Alliance for Affordable Internet (A4AI) uses the target adopted by the UN Broadband Commission that 1 GB of mobile data should cost no more than 2% of average monthly income. In its analysis of 95 low- and middle-income countries, published in 2021, A4AI found that 1 GB in Africa cost on average 5.8% of monthly income, the highest of any region, with variation

from 0.5% in Egypt to 24.4% in the Central African Republic. Seven countries met the target for the first time in that analysis, but 52 of the 95 countries had not met it, leaving almost one billion people in places where basic access was unaffordable (A4AI, 2021). The GSMA adds that more than half of countries in its sample had yet to meet the target and that the cost of entry-level internet-enabled handsets, relative to income, worsened significantly in 2022 after several years of improvement (GSMA, 2023).

For learning, the relevant cost is not the course fee. It is the sum of the fee, the data used, the device, electricity, and the time that could otherwise be spent earning. A video-based course of ten hours may consume several gigabytes of data. At 5.8% of monthly income per gigabyte, such a course can cost more than the average learner can pay without displacing other spending. The point is a general one about material access (van Deursen & van Dijk, 2019): where the continuing costs of use are high relative to income, nominal availability does not produce use.

4.3 Kenya: high subscription counts, uncertain individual access

Kenya is more connected than the regional average, but headline statistics mislead. The Communications Authority reported 68 million active mobile subscriptions at the end of March 2024, a penetration rate of 132.1%, with 51.3 million mobile data subscriptions, of which 72.6% were on mobile broadband, and 38.7 million mobile money subscriptions (reported in ITWeb Africa, 2024). Subscription counts above 100% reflect people who hold more than one SIM card. They should not be read as individual access, and they say nothing about the quality of connection, the cost of data to the user, or whether the person uses the connection for learning.

Kenya's experience with mobile money demonstrates what happens when a locally relevant service meets high mobile penetration. Suri and Jack (2016) estimated that access to the M-PESA mobile money system increased per capita consumption and lifted about 194,000 households, 2% of Kenyan households, out of poverty, with larger effects for female-headed households. The relevance to learning is not that digital learning will have the same effect. It is that in Kenya a service that works on basic handsets, fits how people actually transact and builds on existing channels can reach large numbers of low-income users, including women, at scale.

4.4 Schools and institutions

Schools are in a similar position. The 2023 Global Education Monitoring Report estimated that globally only 40% of primary, 50% of lower secondary and 65% of upper secondary schools were connected to the internet. It reported that almost 9% of the global population lacked electricity in 2021 and that one in four primary schools had no electricity, and that at least half a billion students, 31% of the world's learners and 72% of the poorest, could not be reached by remote learning during the pandemic (UNESCO, 2023). Kenya's National Digital Masterplan

2022 to 2032 sets targets to extend fibre to 40,000 public schools and learning institutions, create 25,000 youth hotspots and establish 1,450 digital innovation hubs (ITWeb Africa, 2022). The targets signal political commitment and a direction of travel. They also imply that, in the medium term, a provider cannot assume that a learner's school or college will be online, or that the learner will have a private device at home.

5. Gate 3: use, or who can operate the service

5.1 Gender

Averages conceal who is excluded. The GSMA's Mobile Gender Gap Report 2024 found that in Sub-Saharan Africa in 2023 women were 32% less likely than men to use mobile internet, that 36% of women used it, and that about 202 million women in the region did not use it. The gap had narrowed from 34% in 2017, the first narrowing in five years (GSMA, 2024b, as reported by Connecting Africa, 2024b). A learning service whose usage data are not disaggregated by sex will systematically overstate whom it reaches, because the people it counts are disproportionately men.

The reasons are plural, and they may differ from the reasons for the income gap. Plausible candidates include less control over household budgets and devices, heavier unpaid care duties that reduce time for study, and safety and social-norm barriers to using phones. The sources reviewed here document the size of the gap and not the relative weight of these causes, and the paper does not attempt to rank them. The design implication is nonetheless clear: a programme that wants to reach women must be designed around their actual channels, schedules and risks, and must measure outcomes by sex from the start. Mobile money offers an instructive Kenyan parallel, since Suri and Jack (2016) found effects larger for women-headed households, which suggests that services built around how women actually manage money can reach them.

5.2 Disability

Disability produces a similar and often larger gap. The 2020 Global Education Monitoring Report, devoted to inclusion, found that children with disabilities make up an estimated 15% of the out-of-school population in the 14 countries with suitable data, and that children with sensory, physical or intellectual disabilities are 2.5 times more likely than their peers never to have attended school. Laws in 25% of countries provide for education in separate settings, 25% of teachers in 48 education systems report a high need for professional development on teaching students with special needs, and only 41 countries recognise sign language as an official language (UNESCO, 2020). On assistive technology, the WHO and UNICEF estimated in 2022 that more than 2.5 billion people need at least one assistive product, that nearly one billion are denied access, and that access in some low- and middle-income countries is as low

as 3% of need, against about 90% in a sample of wealthy countries. They projected that the number in need would reach about 3.5 billion by 2050 (WHO & UNICEF, 2022).

The normative framework is not in doubt. Article 24 of the UN Convention on the Rights of Persons with Disabilities requires an inclusive education system at all levels (United Nations, 2006). The Constitution of Kenya guarantees the right to education and entitles persons with disabilities to access educational institutions and facilities integrated into society (Republic of Kenya, 2010, Arts. 43 and 54), and the Persons with Disabilities Act of 2003 provides statutory support (Republic of Kenya, 2003). What is less settled is the technical standard to which digital learning should be held. Two are widely used. The Web Content Accessibility Guidelines 2.2 set testable criteria for perceivable, operable, understandable and robust web content (W3C, 2023). Universal Design for Learning offers a pedagogical complement, calling for multiple means of engagement, representation, and action and expression (CAST, 2018; Meyer et al., 2014). A provider that does not state which standard it meets, and does not test against it, cannot claim to be accessible, and funders are entitled to ask for the test results.

5.3 Language, skills and the problem of free access

Free or low-cost access does not by itself close gaps in use. The best-documented illustration comes from massive open online courses. Hansen and Reich (2015) analysed demographic data from 68 free courses on HarvardX and MITx in 2012 to 2014 and found that young students who enrolled lived in neighbourhoods with a median income 38% higher than typical American neighbourhoods, and that teenagers with a college-educated parent had nearly twice the odds of finishing compared with those whose parents had not completed college. The authors concluded that open online courses were reproducing, and in some respects widening, existing advantage. The finding concerns the United States, but its mechanism, that people with more education and resources are better placed to use free resources, applies wherever access is formally open.

The same research programme shows that the gap can be narrowed by design. Kizilcec and colleagues (2017) reported that, of more than 25 million people who enrolled in MOOCs between 2012 and 2015, 39% were from developing countries, and that brief randomised writing interventions at the start of a course, one on the relevance of the course to learners' values and one on social belonging, in some cases roughly doubled persistence among learners from less-developed countries and often eliminated the achievement gap between them and learners from developed countries. The point for providers is that exclusion at the use gate is partly psychological and social, and that small, inexpensive design features can matter as much as technical ones.

Language is a further use barrier that deserves separate treatment, and is taken up in the third paper in this series. Here it suffices to note that a learning service built in a language the learner does not use fluently fails at gate 3 whatever its price and speed, and that the 2016

UNESCO policy paper estimated that 40% of the world's population does not have access to education in a language they speak or understand (UNESCO, 2016).

6. Gate 4: access to technology is not access to learning

6.1 Devices without instruction

The evidence that devices alone do not produce learning is consistent. The 2023 Global Education Monitoring Report records that Peru distributed more than one million laptops with no positive effect on learning, partly because attention was given to devices rather than to their pedagogical integration. It also reports a meta-analysis across 14 countries finding a small negative effect of student mobile phone use on outcomes, larger at university level, and that only about 10% of 15-year-olds in more than 50 education systems used digital devices for more than an hour a week in mathematics and science lessons in 2018 (UNESCO, 2023). Escueta, Nickow, Oreopoulos and Quan (2020), in a review of experimental research, concluded that technology can help, and in some cases hinder, learning, and that the effects depend on how it is used.

A second strand of evidence concerns inputs without incentives or instruction more generally. Mbiti and colleagues (2019) randomised interventions across 350 Tanzanian schools and found that unconditional school grants had no measurable effect on test scores after two years, that teacher incentives tied to student performance had some positive effect, and that the two combined had significant effects larger than the sum of the separate effects. The authors interpret this as complementarity between inputs and the incentives that govern how inputs are used. The result is not about technology, but it has a parallel: funding connectivity and devices without funding the instruction and incentives that govern their use may achieve little.

6.2 What works: targeted instruction

The more specific finding is that technology helps when it supports instruction matched to the learner's level. Banerjee, Cole, Duflo and Linden (2007) found in two randomised experiments in India that a computer-assisted learning programme raised mathematics scores by 0.35 standard deviations in the first year and 0.47 in the second, that remedial teaching by community-based tutors also raised scores, and that classmates who benefited only from smaller classes showed no gains. The gains persisted after the programmes ended. Muralidharan, Singh and Ganimian (2019) evaluated a personalised, technology-aided after-school programme in urban India and found, after about 4.5 months, gains of 0.36 standard deviations in mathematics and 0.22 in Hindi for students who won a lottery for a place. They attributed the effect to instruction matched to each student's level, which mattered because

the control group's weaker students learned at a rate close to zero. In both studies, technology served a pedagogical purpose, targeted instruction, rather than substituting for one.

Reviews of the wider literature converge. Evans and Popova (2016) found that six systematic reviews reached divergent conclusions largely because they included different studies, and that two types of programme were recommended consistently: instruction tailored to students' learning levels, delivered by teachers or adaptive software, and individualised, repeated teacher training tied to a specific task or tool. Conn (2017), in a meta-analysis of 56 articles covering 66 experiments and quasi-experiments in Sub-Saharan Africa, found that programmes changing teacher pedagogy or classroom instruction had effect sizes about 0.30 standard deviations larger than all other programme types combined. McEwan (2015), in a meta-analysis of 77 randomised experiments with 111 treatment arms in primary schools of developing countries, provides a wider base for comparing intervention types. The implication for access is direct. Reaching a learner with a device and a connection and no good instruction reaches her and then fails her at the last gate.

6.3 Low-technology designs

Low-technology designs can work where connectivity is constrained. Angrist, Bergman and Matsheng (2022) ran a randomised trial in Botswana of SMS messages and phone calls to parents during school closures and found that the combined treatment improved learning by 0.12 standard deviations at low cost. Aker, Ksoll and Lybbert (2012) evaluated an adult education programme in Niger in which some students were taught to use basic mobile phones, and found that students in the treated villages scored about 0.19 to 0.26 standard deviations higher than students in conventional classes, with the mathematics advantage still present seven months after classes ended. Angrist and colleagues' result is modest in size. Both are important in kind. A service that works over basic voice and text reaches learners whom a video platform does not, and at a cost within reach of many sponsors. Figure 3 collects the reported effects from these and other trials to show orders of magnitude.

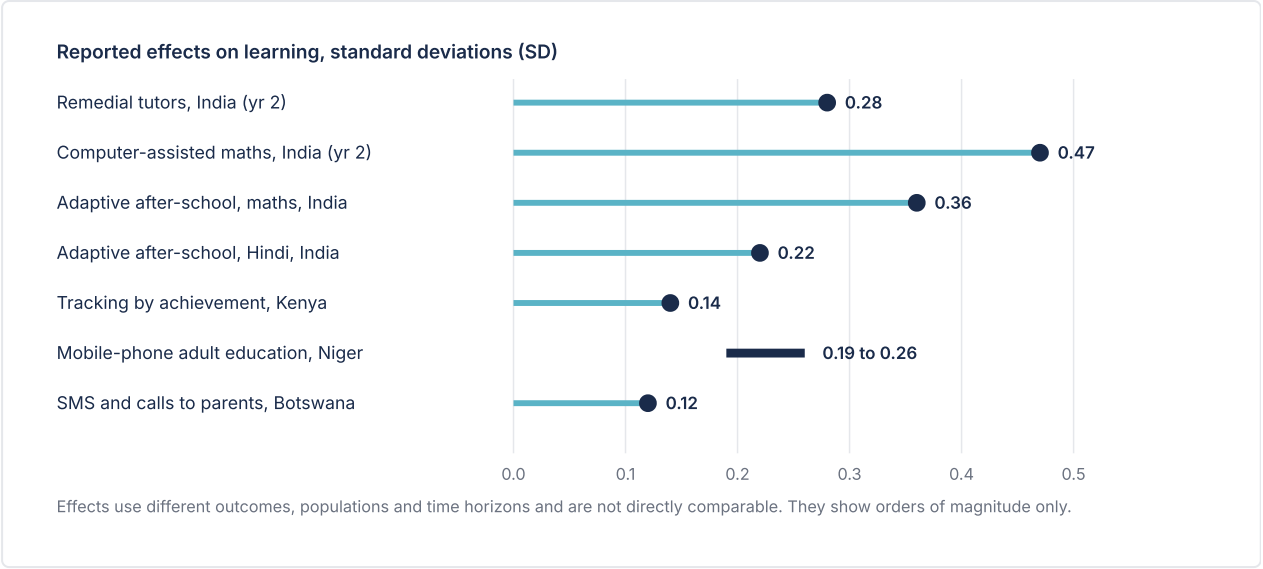


Figure 3. Reported learning effects, in standard deviations, from selected randomised evaluations cited in this paper. Outcomes, populations and time horizons differ, so the figure shows orders of magnitude and not a ranking. Sources: Banerjee et al. (2007); Muralidharan et al. (2019); Duflo et al. (2011); Aker et al. (2012); Angrist et al. (2022).

Two cautions apply. First, nearly all the cited effects come from small-scale or pilot-scale studies run by researchers with strong implementation support, and effects often shrink at scale. Second, effects are measured on narrow outcomes, mostly basic literacy and numeracy, in school-age or adult basic education populations. They are not evidence about advanced or workplace skills, where most digital course providers operate. The evidence base for those populations in Africa is thin, which is an argument for evaluating one's own programmes, the subject of the second paper in this series.

6.4 Scale, context and external validity

A review that draws conclusions for Kenya from trials in India, Niger, Botswana and Tanzania must say how far results travel. Pritchett and Sandefur (2015) examined this directly, using microcredit, migration and education policies, and found that by mean-squared error, non-experimental evidence from the same context performed better than experimental evidence from a different one, and that replication of experiments narrowed but may not reverse this advantage. Their conclusion is not that experiments are uninformative. It is that a randomised result from elsewhere is a prior that must be updated with local evidence, and that decision makers should weight context heavily.

Two implications follow for access. First, the effect sizes in Figure 3 should be treated as plausible orders of magnitude for Kenya, not as forecasts. A provider who cites the Indian adaptive-learning results to claim that its own programme will produce 0.3 standard deviations is making an unsupported inference. Second, local evidence is not a luxury. The most valuable evidence for a Kenyan provider is a well-designed evaluation of its own programme with its

own learners, even if that evaluation sits low on the evidence ladder. A before-and-after measure with a validated instrument in the right population can be more informative for local decisions than a randomised trial in another country.

7. Who pays: the economics of payer-independent access

7.1 The financing problem

The evidence reviewed so far establishes that access fails because of conditions the learner cannot change: the price of data, the language of the service, the absence of accessibility features and the quality of instruction. The corollary is that the learner should not be asked to pay for removing them. If the person who benefits cannot pay, someone else must, and a provider who wants access to be independent of ability to pay must say who that someone is and show that the money reached the learner.

Three funding models recur. In the first, the learner pays a price low enough to be affordable, sometimes with a free tier. This is simple and scalable but excludes the poorest, and the cost of data and devices means that a low fee does not make the service affordable. In the second, an institution pays on behalf of its members: a university, employer, government programme or non-profit. This can fund quality and accessibility, but can exclude people outside institutions. In the third, a sponsor, such as a foundation, development agency or corporate donor, underwrites participation by eligible learners. This is the model that most directly decouples access from ability to pay, and it is also the most dependent on the sponsor's priorities and time horizon. A mature provider is likely to use all three, with cross-subsidy from paying institutional clients to sponsor-funded learners.

7.2 Open licensing and the cost of ownership

Open licensing changes the economics of access by removing the licence fee and by letting institutions host, adapt and share content. The UNESCO Recommendation on Open Educational Resources, adopted unanimously by the 193 member states in November 2019, encourages the creation, adaptation and sharing of openly licensed educational materials (UNESCO, 2019). The Digital Public Goods Standard lists open licensing, clear ownership, platform independence, documentation, a mechanism for extracting data and adherence to privacy laws among its baseline indicators (Digital Public Goods Alliance, n.d.). These instruments do not remove the cost of maintenance, hosting or support, and open materials can be neglected. They do reduce the structural cost of access and the risk that an institution is locked into a vendor.

The Global Education Monitoring Report of 2023 recommends that governments pilot technology in ways that reflect total cost of ownership and establish transparency in spending and contracts with private companies (UNESCO, 2023). A provider that publishes the full cost per learner, including data, devices and facilitation, and the share of learners reached on sponsored terms, makes it possible for funders to compare access models on a common basis. Few do so, and the evidence on the cost per completing learner in Kenya is, to the authors' knowledge, sparse.

7.3 Paying for quality: cost-effectiveness

A common instinct in access programmes is to minimise cost per learner. The evidence suggests that this can be the wrong objective. Piper, Zuilkowski, Dubeck, Jepkemei and King (2018), analysing the Kenyan PRIMR randomised trial, compared combinations of teacher professional development and coaching, student textbooks, and structured teachers' guides, and found that the most expensive combination was also the most cost-effective because its added impact on learning outweighed its added cost. The finding is from early-grade literacy, but the principle generalises: the relevant denominator is not the number of learners reached but the amount of learning produced per unit of money, and a cheaper programme that produces little learning is not cheap.

For sponsors of access this implies two things. A sponsor that pays for connection and a licence but not for facilitation or instruction may be paying for the part of the system that does least. And a provider who reports only the cost per enrolled learner invites the comparison that favours the thinnest offer. Reporting the cost per completing learner and, where possible, the cost per unit of measured learning gain, aligns the provider's incentive with the funder's.

8. Propositions

The framework and evidence imply propositions that can be tested. They are stated as directional claims and are intended as a research agenda.

Table 1. Propositions derived from the conditions-of-access framework

No.	Proposition	Basis in the evidence reviewed
P1	In Kenya, the usage gap among covered adults will be larger than the coverage gap, and will be explained more by cost and skills than by the absence of a signal.	GSMA (2024): usage gap 60%, coverage gap 13% in the region; A4AI (2021); GSMA (2023)
P2	A learning programme that meets the learner's data and device costs will show higher completion among low-income learners than the same programme at the same fee without such support.	A4AI (2021); van Deursen & van Dijk (2019); Hansen & Reich (2015)
P3	Disaggregated reporting by sex and disability will reveal larger exclusion than aggregate usage statistics, and design changes aimed at the excluded group will narrow it.	GSMA (2024b); UNESCO (2020); WHO & UNICEF (2022); Kizilcec et al. (2017)
P4	Programmes built to WCAG 2.2 and UDL standards will show smaller differences in completion between learners with and without disabilities than programmes that are not.	UNESCO (2020); W3C (2023); CAST (2018)
P5	Learning gains per learner will be larger where digital content is paired with targeted instruction or facilitation than where it is delivered alone.	Banerjee et al. (2007); Muralidharan et al. (2019); Escueta et al. (2020); Mbiti et al. (2019)
P6	Low-bandwidth, text- and voice-based delivery will reach a larger share of low-income and rural learners per unit cost than video-based delivery, with learning gains that are smaller per learner but not per dollar.	Angrist et al. (2022); Aker et al. (2012); A4AI (2021)

P2 to P6 are untested in Kenya for adult and workforce learning to the authors' knowledge.

9. Design requirements for payer-independent access

The preceding sections imply that a provider committed to access must treat each gate as a design constraint and each barrier as a cost to be borne by the provider or its sponsors, not by

the learner. We derive seven requirements. Table 2 maps each barrier to the evidence and to the response it implies, and Figure 4 shows which gate each requirement protects.

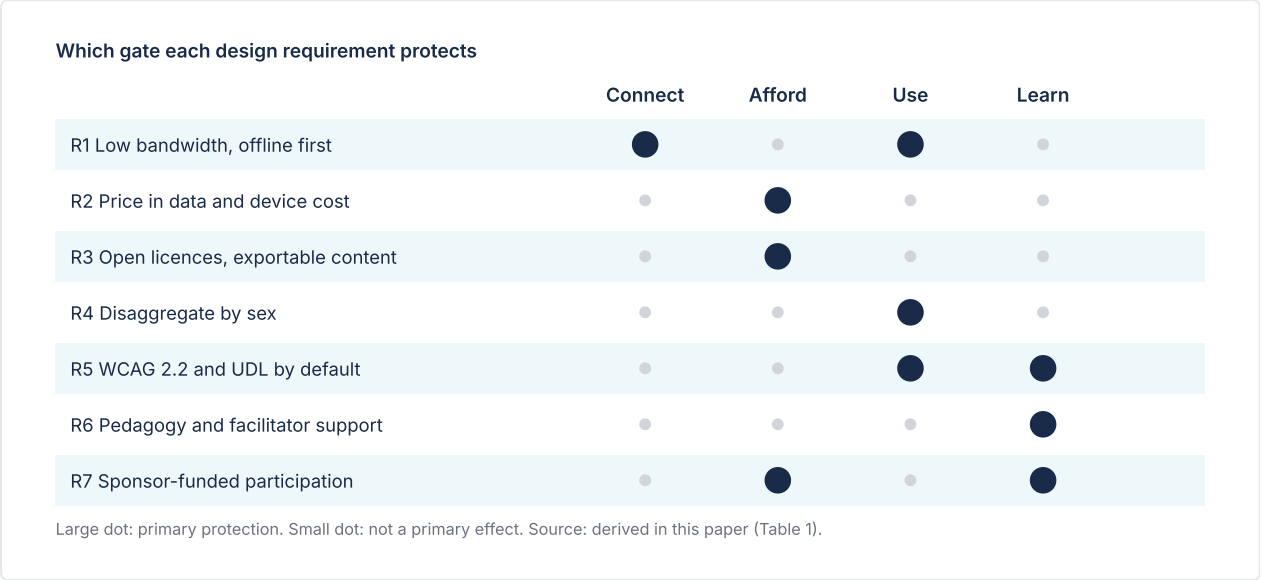


Figure 4. Mapping of the seven design requirements (R1 to R7) to the gates they principally protect. Several requirements protect more than one gate; the large dot marks the principal effect.

Table 2. Barriers to access, supporting evidence and design responses

Barrier	Evidence	Design response
Most people under coverage do not use mobile internet	60% usage gap; 13% coverage gap (GSMA, 2024)	R1. Design for low bandwidth and offline use first; treat a good connection as an enhancement
Data and devices cost more than the target	1 GB at 5.8% of monthly income in Africa against a 2% target (A4AI, 2021); handset costs worsened (GSMA, 2023)	R2. Count the learner's data and device cost as part of price; use mobile-first, lightweight pages and portable packages
Licences and platforms that lock content in	UNESCO Recommendation on Open Educational Resources (2019); Digital Public Goods Standard	R3. Prefer open licences, open-source components and exportable content so that institutions can host and adapt it without a fee
Women offline	32% less likely to use mobile internet; 36% of women online (GSMA, 2024b)	R4. Disaggregate all usage and outcome data by sex; design delivery around women's actual channels and schedules
Disability and assistive technology gaps	2.5 times more likely never to attend school; access to assistive products as low as 3% in some low- and middle-income countries (UNESCO, 2020; WHO & UNICEF, 2022)	R5. Build to WCAG 2.2 and UDL by default and publish test results
Devices without instruction produce no gain	Peru laptops; targeted instruction effects (UNESCO, 2023; Banerjee et al., 2007; Muralidharan et al., 2019)	R6. Build learning design, practice and feedback into the product, and pair it with facilitator support where possible

Barrier	Evidence	Design response
The payer is not the learner	Learning poverty of about 70% (World Bank et al., 2022) alongside data costs of 5.8% of monthly income (A4AI, 2021)	R7. Fund access through sponsors, institutions and cross-subsidy, and report how many learners participated on sponsored terms

Three of these requirements deserve elaboration. R1 reflects the finding that the principal gap is use and not coverage. Designing for low bandwidth does not mean designing for poor quality. It means that the course should degrade gracefully: text and audio first, with compressed images, optional video, and downloadable packages that work offline, so that a learner on a shared handset with intermittent data can complete the course. Content packages that follow open e-learning standards can run on any compliant learning management system and can be distributed through local institutions, which also supports R3.

R5 reflects the observation that accessibility is rarely tested in practice. The standards exist, and testable criteria are available. A provider that states which standard it meets and publishes the results of automated and manual tests, including tests with users who rely on assistive technology, converts a general commitment into one that can be checked. R7 reflects the financing argument in Section 7: if the learner cannot pay, the provider's honest statement of who does is part of the access claim.

10. Reflexive application: Instracta eLearning Solutions

10.1 Purpose and caution

To test whether the framework can be applied, this section applies it to the provider with which the first author is associated. The exercise is reflexive, and it carries the risk of self-congratulation. It is therefore limited to stating, for each requirement, what the provider has in place, what it does not, and what evidence is currently available. It is not an evaluation and does not claim that the provider's programmes improve learning, which is the subject of the second paper in this series.

10.2 Assessment against the requirements

Instracta is a Kenyan learning provider based in Nairobi. Its course authoring tool exports standards-based packages that can run in other learning management systems, which supports R1 and R3, although the provider has not yet published bandwidth benchmarks. Its Moodle theme is released as open source, which supports R3. Its learning management system is in preparation and planned to be built on Moodle, an open-source platform. An accessibility plugin intended to bring a comprehensive set of features to Moodle is planned and not yet released, so R5 is a commitment and not yet a record. Sponsor-funded participation, R7, is what the Instracta Reach initiative is intended to deliver, and it depends on funding that has not yet been secured. R2, R4 and R6 are design intentions for which the provider has no published data, and the evaluation framework that would generate them is set out in the second paper.

Table 3. Provider status against the design requirements (as at October 2026)

Requirement	In place	Not yet in place
R1 Low bandwidth, offline first	Standards-based export from the authoring tool	Published bandwidth and device benchmarks
R2 Price includes data and device cost	Intention stated	Cost-per-learner analysis including data and devices
R3 Open licences and exportable content	Open-source Moodle theme; exportable packages	Open licence policy for all content
R4 Disaggregation by sex	Intention stated	Disaggregated reporting for any cohort
R5 WCAG 2.2 and UDL by default	Planned plugin; commitment	Released plugin; published test results
R6 Pedagogy and facilitator support	Instructional design practice	Evaluated learning gains
R7 Sponsor-funded participation	Initiative designed	Secured sponsor funding; participation data

10.3 What the exercise shows

Two observations follow. First, the framework is operational: it produced a short, checkable list of gaps. Second, most of the gaps are in measurement and publication, not in intention. This is common among small providers and is the reason that Instracta has adopted the evidence ladder described in the second paper. The reflexive application also illustrates the framework's main limitation, which is that it describes conditions and does not weigh them: it does not say which requirement matters most for a given population, and that is an empirical question.

11. Discussion and implications

11.1 For providers

For providers, the most useful result is a change in the unit of analysis. The unit is not the platform or the average user but the learner at the margin. A provider who asks, for each programme, which gate is losing the learners who most need it, will make different choices from one who counts registrations. It will choose a lighter format over a richer one, invest in offline packaging before interactivity, publish accessibility tests before launching new features, and treat the price of data as part of its price. It will also look for partners, such as universities, training colleges, employers and community organisations, that can provide the facilitation that the evidence suggests makes technology effective.

11.2 For funders

Funders can use the framework as a due-diligence tool. Questions that follow from it include: which gate does this programme address, and which does it assume away? What is the cost per learner who completes, including data and devices? What share of learners are women, people with disabilities, and people outside urban areas, and how do their completion and learning compare? What accessibility standard is met and how was it tested? Who pays for learners who cannot? A provider who can answer these questions with data is easier to fund and easier to hold to account. A provider that cannot should be asked what it will do to find out. Funders can also fund the missing evidence directly: the cost per completing learner in accessible, low-bandwidth delivery is a public-good finding that no single provider has an incentive to produce.

11.3 For policy

For policy makers, the data point to a shift in emphasis from coverage to use. Spectrum and infrastructure remain important, but the largest share of people who are not online are under coverage. Policy instruments that reduce the cost of data and entry-level devices, support digital skills, and require accessibility in publicly funded digital learning address the gates that now bind. Kenya's National Digital Masterplan includes targets for connectivity and hubs (ITWeb Africa, 2022). The framework suggests that these targets be complemented by targets

for use, for example the share of women and of persons with disabilities using publicly funded learning services, and for learning outcomes. The 2023 Global Education Monitoring Report recommends independent bodies to evaluate education technology and national targets for meaningful connectivity (UNESCO, 2023).

11.4 For research

Four questions would most improve the evidence. First, what share of eligible Kenyan learners would complete a low-bandwidth course on a basic smartphone when data costs are met by a sponsor, compared with when they are not (P2)? Second, what is the cost per completing learner of delivering accessible, offline-capable learning in Kenyan languages? Third, how do completion and learning gains differ by sex and disability status, and what design changes narrow the gap (P3 and P4)? Fourth, does facilitator support change learning gains enough to justify its cost (P5)? Each can be tested in pilot programmes with pre-specified outcomes and an independent evaluator, and several could be tested within a single randomised programme, since sponsored and unsponsored arms, and facilitated and unfacilitated arms, can be crossed.

11.5 Illustrative learner vignettes

To show how the framework diagnoses exclusion, consider three composite vignettes. They are constructed for illustration from the patterns in the evidence, not drawn from the records of any programme, and they should not be read as data.

The first learner is a young woman in a peri-urban area of Kenya who holds a basic smartphone and a SIM card with intermittent data. She passes the connect gate. At the afford gate she is repeatedly lost: she buys data in small bundles, a ten-hour video course is out of reach, and the handset's storage cannot hold large downloads. A text-and-audio course that can be downloaded once over a free or sponsored connection would keep her in the programme at no extra cost to her. Her use of the service is also likely to be shaped by the time she can find after domestic and care work, which is why completion by sex should be reported.

The second learner is a man with a visual impairment who uses a screen reader. He may connect and afford a service, and is lost at the use gate if course pages lack text alternatives, if interactive elements cannot be operated from the keyboard, or if assessments are image-based. No pricing change helps him. An accessibility standard that is tested with screen-reader users does. The data on assistive technology access (WHO & UNICEF, 2022) imply that he may also lack the assistive product itself, which no course provider can supply, but which a sponsor might.

The third learner is a facilitator at a community training centre who has a laptop and a stable connection, and who can connect, afford and use a service. She is lost at the learn gate if the

course she delivers has no practice or feedback and she has not been trained to support learners. A programme that pairs the content with training for facilitators, the kind of teacher support that the Kenyan evidence favours (Piper et al., 2018), has a better chance of turning access into learning. The vignettes illustrate that a single price or platform choice cannot serve all three, and that the gates differ in who can open them.

11.6 Measurement: an access dashboard

If access is to be measured at the margin, providers need indicators that follow the gates. We propose a minimal dashboard of eight indicators, each to be reported by sex, disability status, age and location. For connect: the share of enrolled learners who access the service over a mobile connection, and the median bandwidth they experience. For afford: the learner's total data and device cost per completed hour of learning, and the share of learners whose participation was sponsored. For use: the share of learners who start the first lesson within seven days of enrolling, and the accessibility conformance level achieved against WCAG 2.2 as tested by users of assistive technology. For learn: the share who complete, and the mean gain on a pre-specified assessment. None of these indicators requires new technology. Each requires a decision to collect and publish the data, which is where small providers most often fall short.

The dashboard also protects against a subtle risk, which is that a provider optimises a gate that is easy to measure at the expense of one that is not. Enrolment and registration counts are easy; completion by disability status is not. Because funders increasingly ask for reach, providers face an incentive to report reach and not inclusion. A published dashboard makes the trade-off visible and gives funders a basis for rewarding the harder work.

12. Limitations

The paper has limitations that bear on its conclusions. First, it is a narrative review without a registered protocol, and it reports no screening counts. Despite the steps in Section 2, selection bias cannot be excluded. Second, much of the strongest experimental evidence on technology-aided learning comes from India and other regions, and its transfer to Kenya is plausible but untested. The Kenyan randomised evidence cited in the companion papers concerns early-grade literacy and school-based interventions and not adult and workforce learning, where most digital course providers operate. Third, connectivity statistics are mostly modelled or self-reported, change quickly, and in two cases were accessed through secondary reports of the original publication, which are marked in the text. Fourth, several of the effect sizes cited derive from pilots with strong implementation support and may shrink at scale. Fifth, the framework describes conditions and does not weigh them. Sixth, the provider assessment in Section 10 is a self-assessment by an interested party. The propositions in Table 1 are offered so that these limits can be addressed by evidence and not by assertion.

13. Conclusion

The central claim of this paper is that access is a property of the delivery system as a whole, and that it should be measured at the margin, at the learner who is hardest to reach, and not at the average user. The data show a region in which networks are expanding faster than use, in which data and handsets remain expensive relative to income, in which women and people with disabilities are disproportionately excluded, and in which technology produces learning when it carries good instruction and little when it does not. A provider that takes these facts seriously will design for low bandwidth, price in the learner's real costs, build accessibility in from the start, pair content with instruction, and look for sponsors so that a skill does not depend on who can pay. It will also report honestly where it falls short. The conditions-of-access framework, the seven requirements and the six propositions offered here are intended to make that kind of reporting possible, and the claim of access something that can be checked.

Declarations

Declaration of competing interest

The first author is the founder of Instracta eLearning Solutions, whose products and initiatives are discussed in Sections 9 and 10. No other competing interests are declared.

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Declaration of generative AI use

Drafting of this working paper used generative AI assistance (Claude, Anthropic) for literature searching, structuring and drafting. Bibliographic details were checked against publisher or repository records. The authors reviewed the text and are responsible for its content.

Author contributions (CRediT)

Opiyo Michael Ouru: Conceptualization, Methodology, Investigation, Writing: original draft, Writing: review and editing.

Kenneth Goga Riany: Writing: review and editing.

Daisy Chepkoech O'maera: Writing: review and editing.

Data availability

No new data were generated. All sources are listed in the reference list; the search strings used are given in Appendix A.

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Appendix A. Search strings and source types

Searches were conducted in October 2026. The principal strings are listed below. Searches were iterative and results were screened by title and abstract against the criteria in Section 2.3.

Table A1. Principal search strings and source types

Gate	Search string (abridged)	Source types
Connect and afford	mobile internet usage gap coverage gap Sub-Saharan Africa; 1 GB data affordability target; Kenya mobile subscriptions Communications Authority; offline population ITU 2023	GSMA, ITU, A4AI, Communications Authority via secondary reports
Use	mobile gender gap Sub-Saharan Africa; children with disabilities out of school; assistive technology access; MOOC participants socioeconomic status; closing global achievement gaps MOOCs	GSMA, UNESCO GEM, WHO and UNICEF, Science
Learn	technology-aided instruction randomised India; upgrading education with technology; low-tech learning when school is out; mobile phones adult education Niger; learning interventions Sub-Saharan Africa meta-analysis; inputs incentives complementarities Tanzania	AEA journals, NBER, Review of Educational Research, Nature Human Behaviour, repository records
Context and policy	learning poverty 2022 update; Kenya youth not in employment education or training; Kenya National Digital Masterplan; UNESCO Recommendation on Open Educational Resources; Digital Public Goods Standard	World Bank, KNBS via press, UNESCO, DPGA, trade press
Theory	capability approach; multi-level digital divide; material access; ICT4D design-reality gaps	Books and journals; repository records

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